

Telco Customer Churn

EDA, Model Training & Evaluation — Documentation

This document explains the analysis notebook (notebooks/churn_eda_and_model.ipynb): the data audit and cleaning decisions, the exploratory analysis, how the churn model was trained and selected, the evaluation-metric choice and its justification, and the final results. Every number here is recomputed from the dataset with the notebook's fixed random seed (42).

7,043

customers

26.5%

churn rate

0.6305

PR-AUC (holdout)

0.8395

ROC-AUC (holdout)

Contents

- 1 Dataset overview & audit
- 2 Cleaning decisions
- 3 EDA — churn drivers
- 4 EDA — tenure, charges & collinearity
- 5 Feature preparation & model selection
- 6 Evaluation metric — choice & justification
- 7 Holdout results & calibration
- 8 What drives the model
- 9 Reproducibility, artifact & limitations

1. Dataset overview & audit

The file contains 7,043 customers and 21 columns: a customer identifier, 18 categorical attributes (demographics, subscribed services, contract and billing), two numeric attributes (tenure in months, MonthlyCharges), a derived numeric TotalCharges, and the Churn label. 1869 customers (26.54%) churned. The brief deliberately does not say what is wrong with the file, so the notebook opens with a broad audit: dtypes, missing values, duplicates, whitespace, category domains, numeric ranges, and cross-field consistency.

Finding 1 — TotalCharges loads as text.

Eleven values contain a single space instead of a number, which forces the whole column to text. All eleven belong to tenure = 0 customers whose Churn label is “No” — brand-new, never-billed customers. The blank is structural (nothing has been billed yet), not corruption.

Finding 2 — SeniorCitizen is encoded 0/1.

Every other binary column uses the strings Yes/No; SeniorCitizen alone uses integers 0/1. Harmless for a model, inconsistent for analysis.

Null-result audit — checks that came back clean.

Knowing what is not wrong matters as much as what is. Each check below found zero problems; the file is the pristine classic Telco dataset apart from the two findings above.

audit check	problems found
duplicate rows / duplicate customer IDs	0 / 0
stray whitespace or casing outside TotalCharges	0
negative or impossible numeric values	0
service-flag inconsistencies (e.g. 'No internet service' vs InternetService)	0
TotalCharges far from tenure × MonthlyCharges (ratio < 0.4 or > 2)	0

2. Cleaning decisions

Two fixes, chosen to be minimal and fully documented. The cleaning code asserts the structure it relies on (exactly 11 blanks, all tenure = 0, all non-churned), so if the input file ever changes shape the pipeline fails loudly instead of silently mis-cleaning.

- **Impute TotalCharges = 0.0 and keep the 11 rows.** TotalCharges is the amount billed so far for a never-billed customer — a fact, not a statistical guess. Keeping the rows means every count computed from the cleaned data matches the source file exactly (7,043). The alternative — dropping 11 of 7,043 rows (0.16%) — is defensible for pure modelling but silently desynchronises counts from the shipped file.
- **Map SeniorCitizen 0/1 to No/Yes.** consistency with every other binary column.
- **customerID kept as a lookup column, excluded from model features.** It's a unique identifier, not a feature.
- **Dtypes after cleaning:** categoricals, int64 tenure, float64 charges; no missing values anywhere. A validation routine re-checks row count, category domains, value ranges and service-flag consistency on every load, and the same checks run as the project's data test-suite.

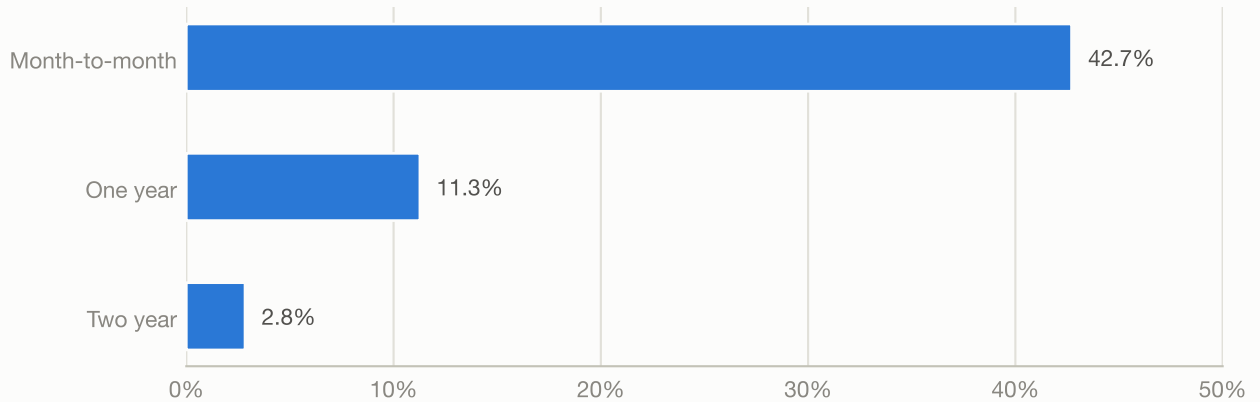
Issues log (as recorded in the notebook)

- TotalCharges was read as text: 11 rows contain a single space instead of a number. All 11 are tenure=0 customers (never billed yet, all Churn=No). Fixed by imputing 0.0 — semantically the true amount billed so far — and keeping the rows so counts always match the source file.
- SeniorCitizen was encoded 0/1 while every other binary column uses Yes/No. Normalized to Yes/No for consistency.

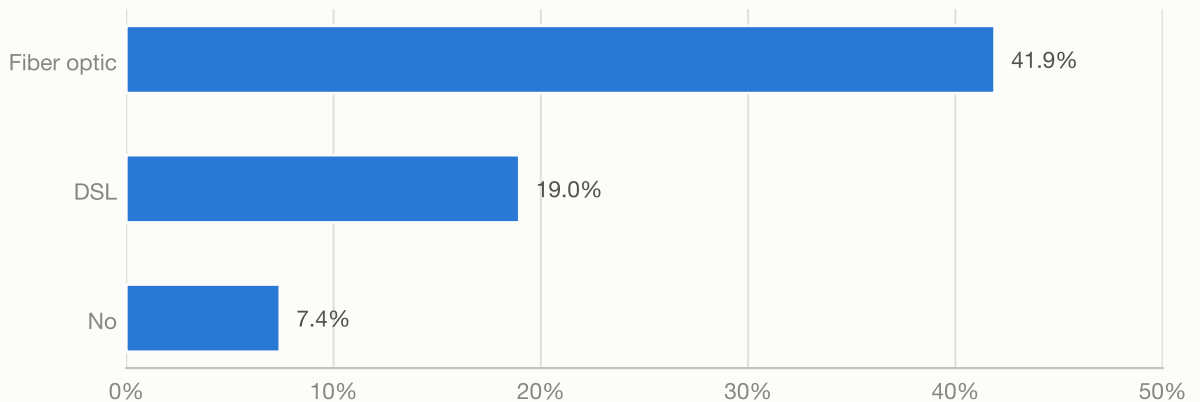
3. EDA — churn drivers

Churn is 26.54% overall (1,869 of 7,043) — a roughly 1 : 2.8 class imbalance that shapes the metric choice in section 6. The strongest visible driver is contract type: month-to-month customers churn at 42.7%, one-year at 11.3%, two-year at 2.8%. Fiber-optic internet and electronic-check payment also stand out; the model's coefficients in section 8 confirm all three.

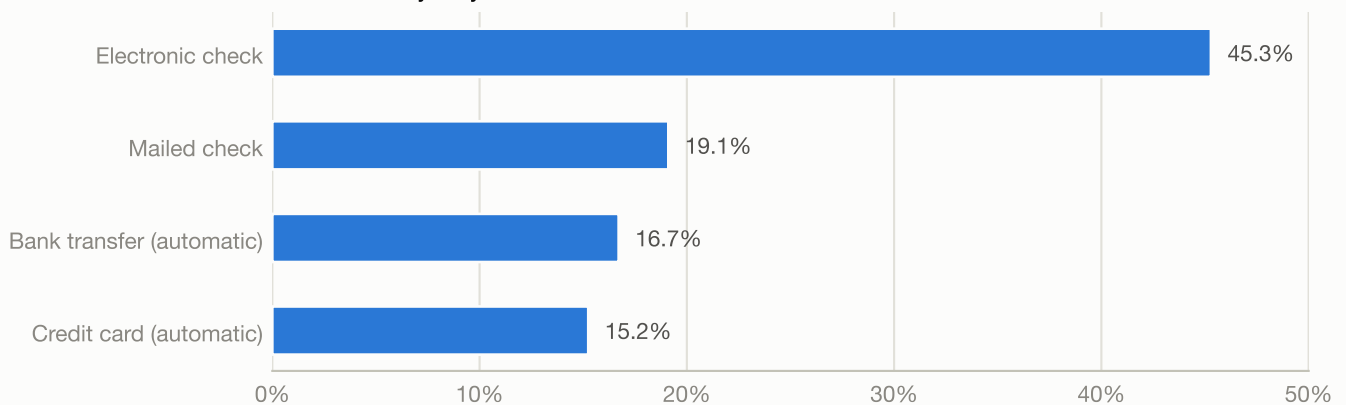
churn rate by Contract



churn rate by InternetService

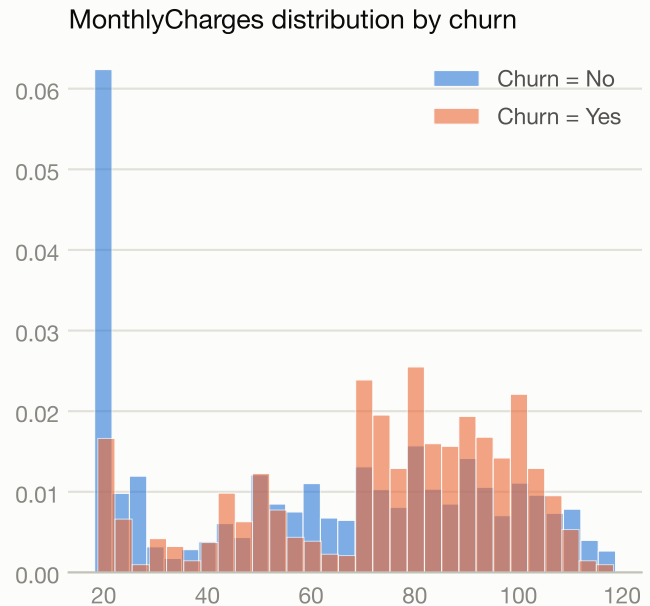
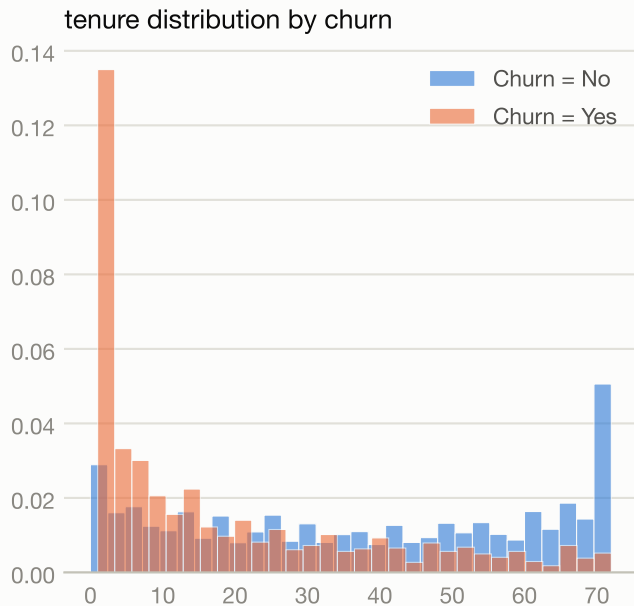


churn rate by PaymentMethod



4. EDA — tenure, charges & collinearity

Churners concentrate at short tenure and high monthly charges: the archetypal at-risk customer is new and paying a lot. Densities below are normalised per group so the smaller churner class remains comparable.



Collinearity: $\text{TotalCharges} \approx \text{tenure} \times \text{MonthlyCharges}$

The correlation between TotalCharges and the product $\text{tenure} \times \text{MonthlyCharges}$ is 0.9996 — the column is essentially a restatement of the other two. It is therefore excluded from model features (kept in the data for analysis): near-collinearity destabilises logistic-regression coefficients and would muddy the per-customer explanations, and any scenario analysis that changes tenure while holding a stale TotalCharges would describe an impossible customer.

	tenure	MonthlyCharges	TotalCharges
tenure	1.000	0.248	0.826
MonthlyCharges	0.248	1.000	0.651
TotalCharges	0.826	0.651	1.000

5. Feature preparation & model selection

Fifteen categorical columns are one-hot encoded; tenure and MonthlyCharges are standardised. customerID (identifier) and TotalCharges (collinear) are excluded; the target is Churn = Yes. Preprocessing and classifier live in a single scikit-learn Pipeline, so the served model can never drift from the transformations it was trained with.

Protocol

Stratified 80/20 holdout (final report), stratified 5-fold cross-validation on the training split (selection), fixed seed 42. Stratification preserves the 26.5% churn share in every fold.

Candidates (5-fold CV on the training split, recomputed)

model	PR-AUC (5-fold CV)	ROC-AUC	Brier
majority baseline	0.2654 ± 0.0001	0.5000 ± 0.0000	0.1949
logistic regression (C=0.1)	0.6599 ± 0.0189	0.8448 ± 0.0118	0.1355
hist gradient boosting	0.6486 ± 0.0212	0.8355 ± 0.0095	0.1400

Logistic regression beats the gradient-boosted trees on this dataset while being exactly explainable per customer (section 8), so the simpler model wins without a trade-off. The C grid {0.1, 1, 10} was chosen on CV average precision; C = 0.1 (stronger regularisation) won.

Class imbalance — deliberately no class_weight=“balanced”

Reweighting inflates predicted probabilities toward the decision boundary and destroys calibration — and the probability itself is what this model serves. Imbalance is handled by stratified splits, minority-focused ranking metrics, and reporting a percentile alongside each score, not by distorting the probabilities.

6. Evaluation metric — choice & justification

There is no single correct metric; the justification is the deliverable. The reasoning, in order:

- **Rejected: accuracy.** With 28.5% churn, predicting “nobody churns” scores 73.5% while being useless. Any single 0.5-threshold metric inherits a milder version of the same blindness.
- **Primary: PR-AUC (average precision).** How well the minority class we care about is ranked. PR-AUC gives no credit for confidently ranking easy non-churners — unlike ROC-AUC under imbalance — so it is the most honest single number for this task.
- **Secondary: ROC-AUC.** A franca for comparing models on this well-known dataset and as a sanity anchor for the PR-AUC reading.
- **Operational: recall@top-decile, precision@top-decile, lift.** Set a threshold: “if we can call 10% of the base, what share of true churners is on the list, and how much better than random is that?” This ties the metric to the decision the model actually supports.
- **Calibration: Brier score & 10-bin reliability table.** Predicted probabilities to people, so 0.8 must mean roughly 80%. This is also why class reweighting was rejected (section 5).

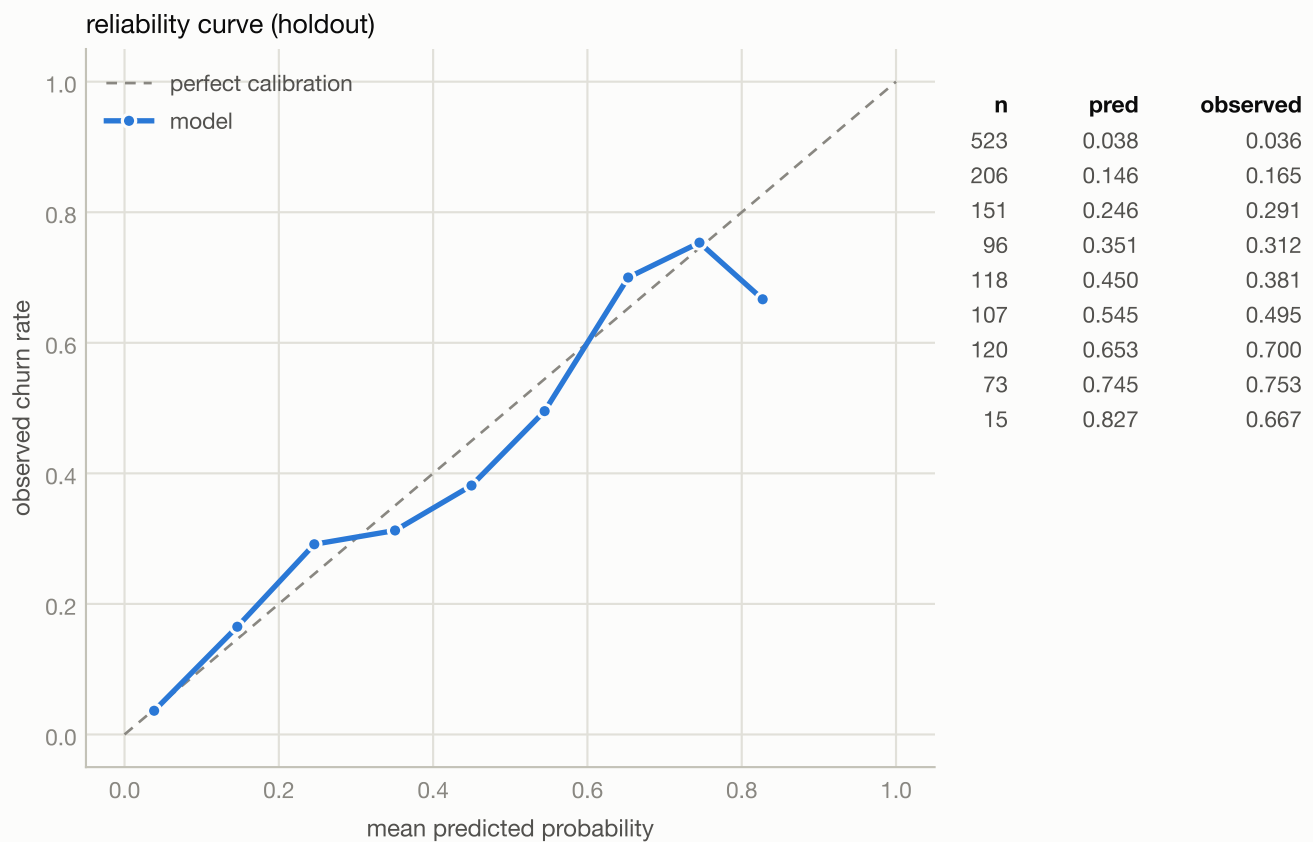
Summary: rank quality for the minority class (PR-AUC) decides model selection; top-decile capture describes operational value under a budget; calibration guarantees the probabilities mean what they say. No threshold is enshrined — the score plus its percentile is the product.

7. Holdout results & calibration

All metrics below describe the exact served artifact (the train-split fit) evaluated on the untouched stratified 20% holdout (1,409 customers).

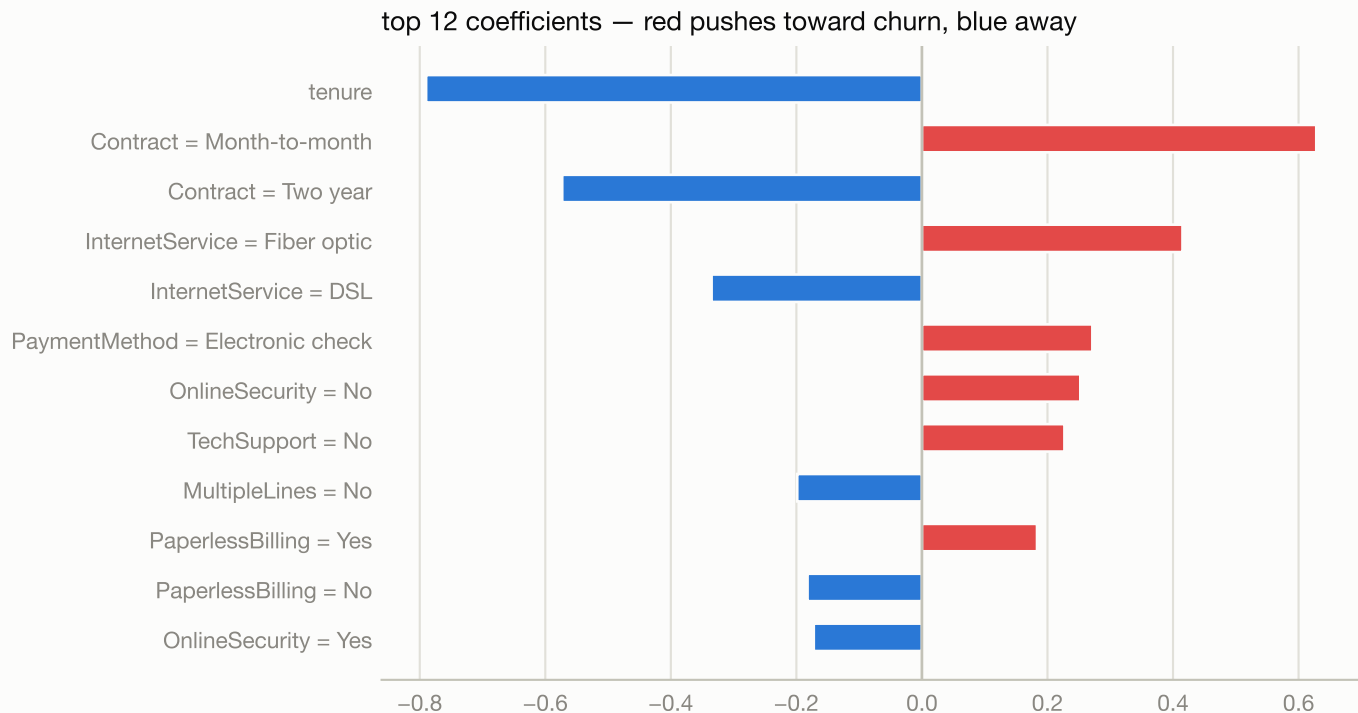
metric	value	reading
PR-AUC	0.6305	vs. 0.2654 for a no-skill ranker (the base rate)
ROC-AUC	0.8395	strong separation for a linear model
Brier score	0.1389	lower is better; 0.195 for always-predict-base-rate
recall@top-10%	0.2754	share of all churners captured in the top decile
precision@top-10%	0.7357	share of the top decile that truly churns
lift@top-10%	2.77	× better than contacting customers at random
churn base rate	0.2654	holdout prevalence (stratified)

Reliability: ten equal-width probability bins, mean predicted probability against observed churn rate. A calibrated model tracks the diagonal — the evidence behind quoting raw probabilities. Bins above 0.8 hold very few customers, so their observed rates are noisy.



8. What drives the model

Largest coefficients of the final logistic regression, in log-odds per one-hot level or per standard deviation. They confirm the EDA: short tenure, month-to-month contracts, fiber-optic internet and electronic-check payment push risk up; long contracts and tenure pull it down.



Per-customer explanation (the notebook's worked example)

For any customer, the contribution of each original feature is its coefficient times the deviation of the customer's transformed value from the training mean, with one-hot terms folded back to their parent column — exact for a linear model. Example: customer 4376-KFVRS scores 0.035. The signed log-odds contributions:

feature	customer value	log-odds contribution
tenure	72	-1.271 (decreases risk)
Contract	Two year	-0.769 (decreases risk)
InternetService	Fiber optic	+0.364 (increases risk)
MonthlyCharges	114.05	+0.265 (increases risk)
OnlineSecurity	Yes	-0.231 (decreases risk)
TechSupport	Yes	-0.199 (decreases risk)

9. Reproducibility, artifact & limitations

Reproduce

- **Notebooks:** churn_eda_and_model.ipynb — fully executed top-to-bottom; every code cell is preceded by a markdown cell explaining what it does and why. Runs locally from the repo or on Google Colab (the first cell falls back to the repository's raw CSV URL).
- **Script:** python -m churn_agent.train retrains the identical pipeline (same seed, same protocol) and rewrites the artifact and metrics.
- **Tests:** data-cleaning invariants and model behaviours run as an offline pytest suite.

Saved artifact (artifacts/model.joblib)

One bundle containing: the fitted Pipeline; feature lists and transformed-feature names; the training-mean vector used for explanations; per-column defaults (median/mode); allowed category values; the sorted training scores (for percentile context); and the holdout metrics (also written to artifacts/metrics.json, reproduced in section 7).

Limitations — stated, not hidden

- **In-sample lookups.** The fitted model is the train-split fit; a score looked up for one of the 80% training customers is in-sample. Holdout metrics describe generalisation; per-customer scores for training members are slightly optimistic.
- **Single snapshot.** This dataset is one point in time — no temporal validation or drift analysis is possible, and tenure acts as both exposure and feature.
- **Linear main effects.** No interaction terms; the gradient-boosting comparison bounds what nonlinearity would buy on this data (it bought nothing).
- **Sparse high-probability bins.** Above 0.8, so calibration there is noisy (see the reliability table).